MONASH University

ETC4500/ETC5450 Advanced R programming

Week 11: Rewriting R code in C++

About me

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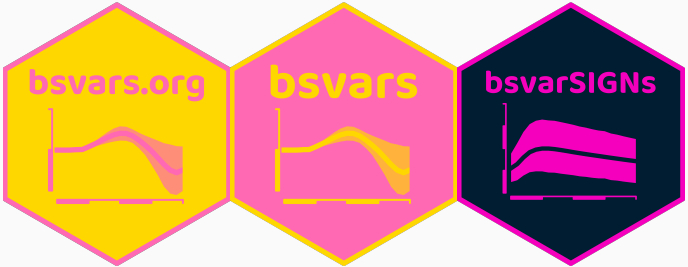
- senior lecturer in econometrics at the unimelb
- econometrician: Bayesian time series analyst
- develops methods for applied macro research
- loves cycling, yoga, books, volunteering, contemporary theatre, music, and arts
- I am nice!

About me

Tomasz Woźniak

- **R** enthusiast and specialised user for 17 years
- associate editor of the R Journal
- author of **R** packages **bsvars** and **bsvarSIGNs**



Motivations

- Compiled code written in **C++** runs much faster than interpreted code in **R**
- Coding in **C++** for **R** applications has always been possible
- It requires:
  - writing **C++** code
  - compiling it, and
  - linking it to **R**
- Difficulties:
  - tedious object-oriented programming
  - necessity of assuring object compatibility
- Benefits are great, but the cost was too high

Motivations

- **Rcpp** is a family of packages by Dirk Eddelbuetel et al. facilitating the application of **C++** in **R**
- An interface for communication between **R** and **C++**
- Greatly simplifies the workflow
- Easier to benefit from the best of the two worlds:
  - **C++** programs are pre-compiled assuring fast computations *perfect for writing functions*
  - **R** code is interpreted and dynamic: *perfect for data analysis*

Objectives for this session

- to facilitate working with **C++** in **R** applications
- to perform a sequence of exercises
- to focus on:
  - basic programming structures
  - functional programming
  - object types: scalars, vectors, matrices, lists, etc.
  - linear algebra
  - statistical distributions

Materials for this session

- Lecture slides
- **C++** scripts:
  - `nicetry.cpp`
  - `nicelr.cpp`
  - `nicelist.cpp`
  - `nicerig2.cpp`

Learning resources

- This session!
- vignettes: for packages **Rcpp** and **RcppArmadillo**
- online resources:
  - **Armadillo** library documentation
  - RcppGallery
  - stackoverflow.com tag:rcpp
- François, R., *Optimizing R Code with Rcpp* on datacamp
- Tsuda, M., *Rcpp for everyone*
- Eddelbuettel, D., *Seamless R and C++ Integration with Rcpp*

The first steps with Rcpp

Consider the following **C++** applications in **R**:

- Define a **C++** function in an **R** script
  - promptly available for fast computations
- Develop a **C++** function in a `.cpp` file
  - perfect for developing, testing, and benchmarking
- Use a function from a `*.cpp` file in **R** computations
  - perfect for elaborate projects
- Develop an **R** package using **C++** code
  - perfect for sharing your work with the community

Define a C++ function in an R script

```
Rcpp::cppFunction('
  DataFrame nicetry (int n) {
    NumericVector v = rnorm(n);
    IntegerVector x = seq_len(n);
    LogicalVector y = v > 0;
    CharacterVector z(n, "nice");
    return DataFrame::create(_["v"] = v, _["x"] = x, _["y"] = y, _["z"] = z);
  }
')
nicetry(2)
```

```
       v x     y    z
1 -0.405 1 FALSE nice
2 -0.807 2 FALSE nice
```

Develop a C++ function in a `nicetry.cpp` file

A `*.cpp` file sample contents:

```
#include <Rcpp.h>
using namespace Rcpp;

// [[Rcpp::export]]
List nicetry (int n) {
  NumericVector v = rnorm(n);
  IntegerVector x = seq_len(n);
  LogicalVector y = v > 0;
  CharacterVector z(n, "nice");
  return List::create(_["v"] = v, _["x"] = x, _["y"] = y, _["z"] = z);
}
/*** R
nicetry(2)
*/
```

Develop a C++ function in a `nicetry.cpp` file

The script includes:

- **Rcpp** library and `namespace` declarations (skip: `Rcpp::`)

```
#include <Rcpp.h>
using namespace Rcpp;
```

- **Rcpp** marker to export the `nicetry` function to R

```
// [[Rcpp::export]]
```

- sample **R** script

```
/*** R
nicetry(2)
*/
```

Develop a C++ function in a `nicetry.cpp` file

The script includes:

- the function definition

```
List nicetry (                      // output type and function name
    int n                           // input type and name
) {
  NumericVector v = rnorm(n);       // define a numeric vector and fill it
  IntegerVector x = seq_len(n);     // define an integer vector as a sequence
  LogicalVector y = v > 0;          // define a logical vector
  CharacterVector z(n, "nice");     // define a character vector
  // return a list with the created vectors
  return List::create(_["v"] = v, _["x"] = x, _["y"] = y, _["z"] = z);
}
```

Develop a C++ function in a `.cpp` file

Your turn!

Develop a **C++** function that creates a `Tx3` matrix with:

- an integer `T` as the only argument
- a constant term column: `NumericVector i(n, 1.0);`
- a linear trend $t - \bar{t}$ column
- a quadratic trend $(t - \bar{t})^2$ column

where t goes from 1 to T, and $\bar{t}$ is the mean of sequence t.

- create `NumericVectors` and assemble as `NumericMatrix`
- use functions `cumsum`, `mean`, `pow`, and `cbind`.

Get some help HERE

Use a function from a `nicelist.cpp` file in R

- `nicelist.cpp` file contents:

```
#include <Rcpp.h>
using namespace Rcpp;

// [[Rcpp::export]]
List nicelist (int n) {
  NumericVector p = rnorm(n);
  NumericVector s(n);
  for (int i=0; i<n; i++) {
    s[i] =  pow(p[i], 2);
  }
  return List::create(_["p"] = p, _["s"] = s);
}
```

Use a function from a `nicelist.cpp` file in R

- **R** script using the function from `nicelist.cpp`:

```r
Rcpp::sourceCpp("nicelist.cpp")
nicelist(3)
```

```
$p
[1] -0.873  1.809 -0.304

$s
[1] 0.7616 3.2726 0.0923
```

Develop a C++ function in a `.cpp` file

Your turn!

Consider a Gaussian random walk:

$$y_t = y_{t-1} + \varepsilon_t, \qquad \varepsilon_t \sim N(0, 1), \qquad y_0 = 0$$

Develop a **C++** function that:

- has an integer T as the only argument
- returns a T-vector with Gaussian random walk

Hint: use functions `rnorm` and `cumsum`.

Some stats with RcppArmadillo

- Data objects from **Rcpp** have limited functionality
- **Armadillo** is a **C++** library for linear algebra that
  - provides a rich set of functions
  - has a simple and intuitive syntax
  - includes fast linear algebra routines, and
  - fast random number generators
  - has fantastic documentation
- **RcppArmadillo** is a simplified interface with **Armadillo**
  - allows seamless integration with **Rcpp**
  - easily passes data between **R** and **C++**

Some stats with RcppArmadillo: IG2 distribution

Sampling random draws from an inverted gamma 2 distribution.

A positive random variable σ^2 following an inverted gamma 2 distribution with positive scale s and shape ν parameters is denoted by:

$$\sigma^2 \sim IG2\,(s, \nu)$$

1. Generate random draw x from $\chi^2(\nu)$
2. Return $\frac{s}{x}$

Some stats with RcppArmadillo: IG2 distribution

Contents of a `nicerig2.cpp` file:

```
#include <RcppArmadillo.h>
// [[Rcpp::depends(RcppArmadillo)]]
using namespace arma;

// [[Rcpp::export]]
vec nicerig2 (
  const int n,
  const double s,
  const double nu
) {
  vec rig2 = s / chi2rnd( nu, n );
  return rig2;
}

/*** R
nicerig2(2, 1, 1)
```

Develop a C++ function in a `.cpp` file

Your turn!

Consider a Gaussian random walk:

$$y_t = y_{t-1} + \varepsilon_t, \qquad \varepsilon_t \sim N(0, 1), \qquad y_0 = 0$$

Develop a **C++** function using **RcppArmadillo** that:

- has an integer T as the only argument
- returns a T-vector of type `vec` with Gaussian random walk

Get some help HERE.

Some stats with RcppArmadillo: linear regression

Contents of a `nicelr.cpp` file:

```
#include <RcppArmadillo.h>
// [[Rcpp::depends(RcppArmadillo)]]
using namespace arma;

// [[Rcpp::export]]
vec nicelr (vec y, mat x) {
  vec beta_hat = solve(x.t() * x, x.t() * y);
  return beta_hat;
}

/*** R
x = cbind(rep(1,5),1:5); y = x %*% c(1,2) + rnorm(5)
nicelr(y, x)
*/
```

Some stats with RcppArmadillo: linear regression

Your turn!

Extend the `nicelr` function to return the covariance:

$$\widehat{Cov}\left[\hat{\beta}\right] = \hat{\sigma}^2 (X'X)^{-1}, \text{ where } \hat{\sigma}^2 = \frac{1}{T}\left(Y - \hat{\beta}X\right)'\left(Y - \hat{\beta}X\right)$$

- don't adjust the arguments
- return `beta_hat` and `cov_beta_hat` in a list

Get some help HERE.

Hint: use functions `inv_sympd` and `.n_elem`.

Some stats with RcppArmadillo: Simulation smoother

Additional resources!

Have a look at my article on *Simulation Smoother using RcppArmadillo* at *Rcpp Gallery*.

An R package with compiled code

Run the following code in **R**:

```
RcppArmadillo::RcppArmadillo.package.skeleton("nicepackage")
```

Note: this function has a different effect if package **pkgKitten** is installed.

An R package with compiled code

- DESCRIPTION includes necessary dependencies

```
Imports: Rcpp (>= 1.0.14)
LinkingTo: Rcpp, RcppArmadillo
```

- NAMESPACE includes dynamic library definition and imports

```
useDynLib(nicepackage, .registration=TRUE)
importFrom(Rcpp, evalCpp)
```

An R package with compiled code

- **C++** code lives in `src/`
  - `src/Makevars` files specify compilation flags
  - `src/Makevars.win` files specify compilation flags for Windows
  - analyse sample `src/*.cpp` file
  - files `src/RcppExports.cpp` and `R/RcppExports.R` are generated automatically by running `Rcpp::compileAttributes()`
  - analyse **R** wrappers to **C++** functions in `R/RcppExports.R`

An R package with compiled code

Your turn!

Create an **R** package with compiled code following the steps from repository donotdespair/15steps2nicepackage

- Read the `README` file
- download file `nicepackage.R`
- follow the instructions in **R**

What's next?

- Keep programming in **C++** for **R** applications
- Reach out for help
- Read the documentation of the **C++** libraries you're about to use
- Study the **Rcpp** family of packages
- Study **openMP** to facilitate parallel computing

What's next?

Rewrite all your code in Rcpp!

Nice!